May Phenotyping Sensors Provide a Potential for Sorting Lines?

The role of deep learning

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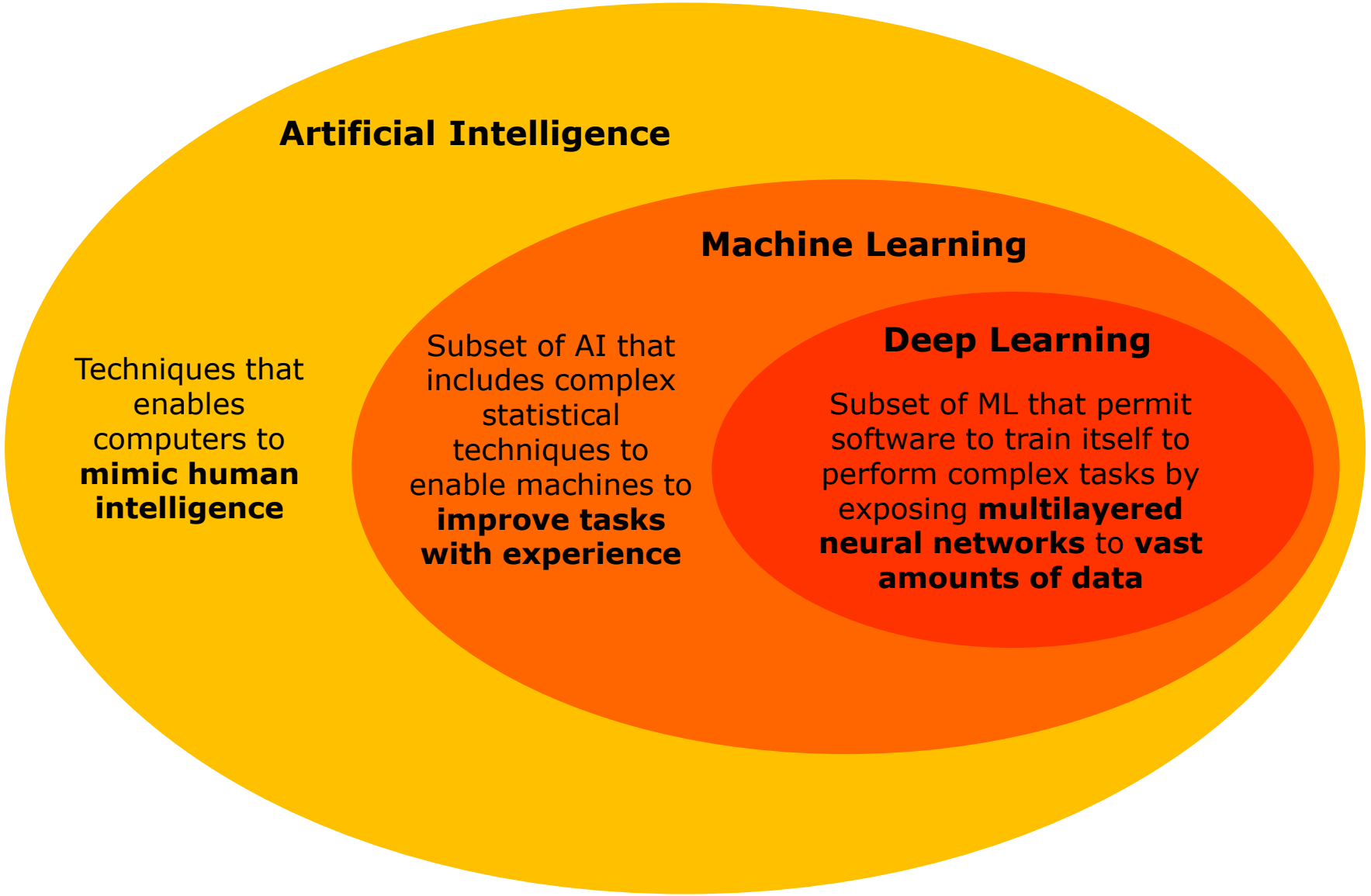
Introduction

- **Phenotyping** is determining the distinctive traits of plants
- **Sorting lines** sort fruit based on visible or invisible quality traits
- **Quality traits** can depend on phenotype (i.e. physical properties) or external factors (i.e. diseases or defects)
- **Sensors** like colour computer vision systems and imaging or point spectral systems, are used to determine these quality traits in sorting lines
- **Computer vision systems** are fully adopted by industry. For spectral devices, more research is needed
- This talk shows a study on the application of hyperspectral imaging combined with a **deep learning algorithm** (autoencoder) to detect **texture changes in persimmon** that can severely depreciate the fruit and short the shelf life

Introduction



Introduction

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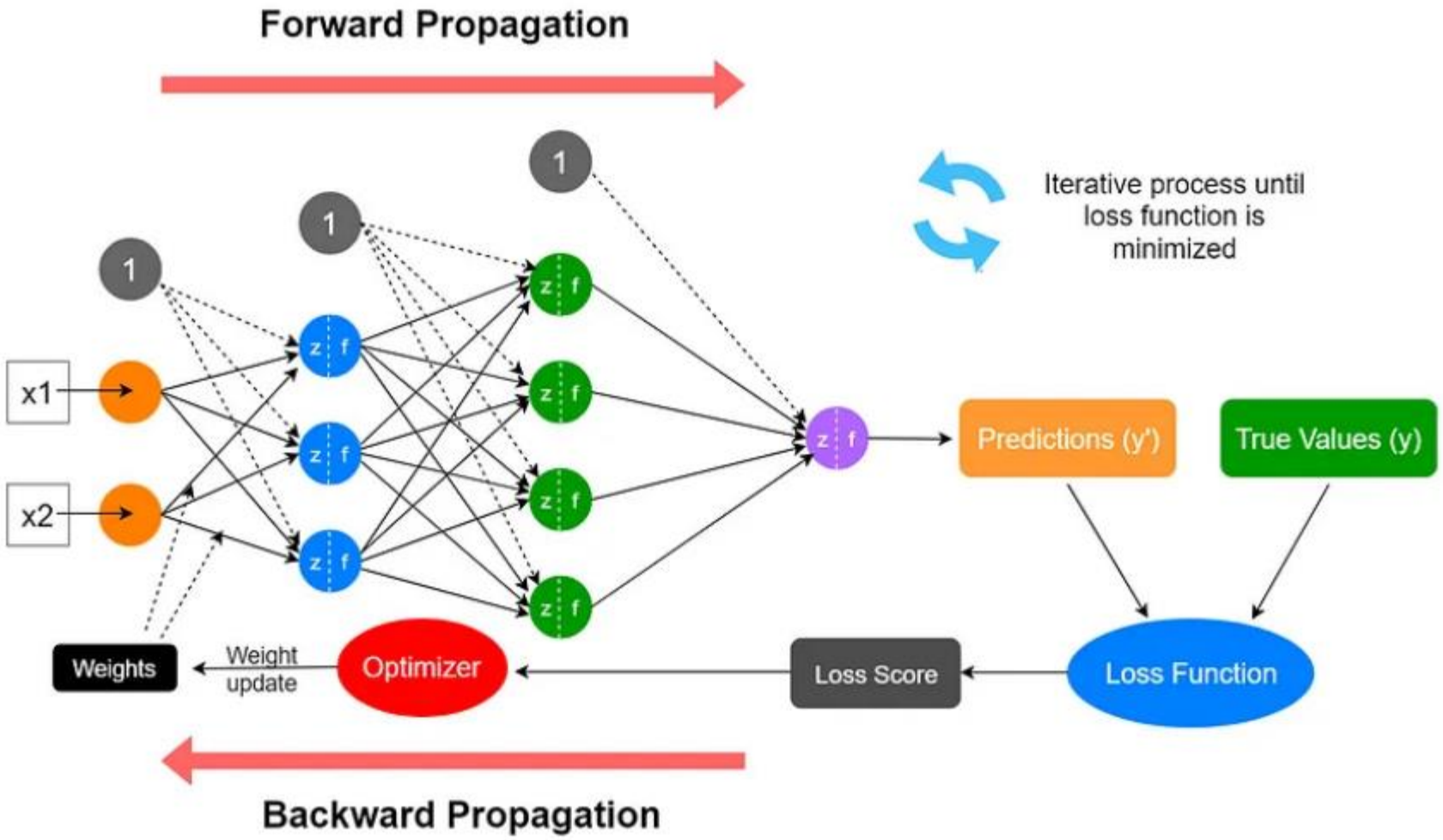
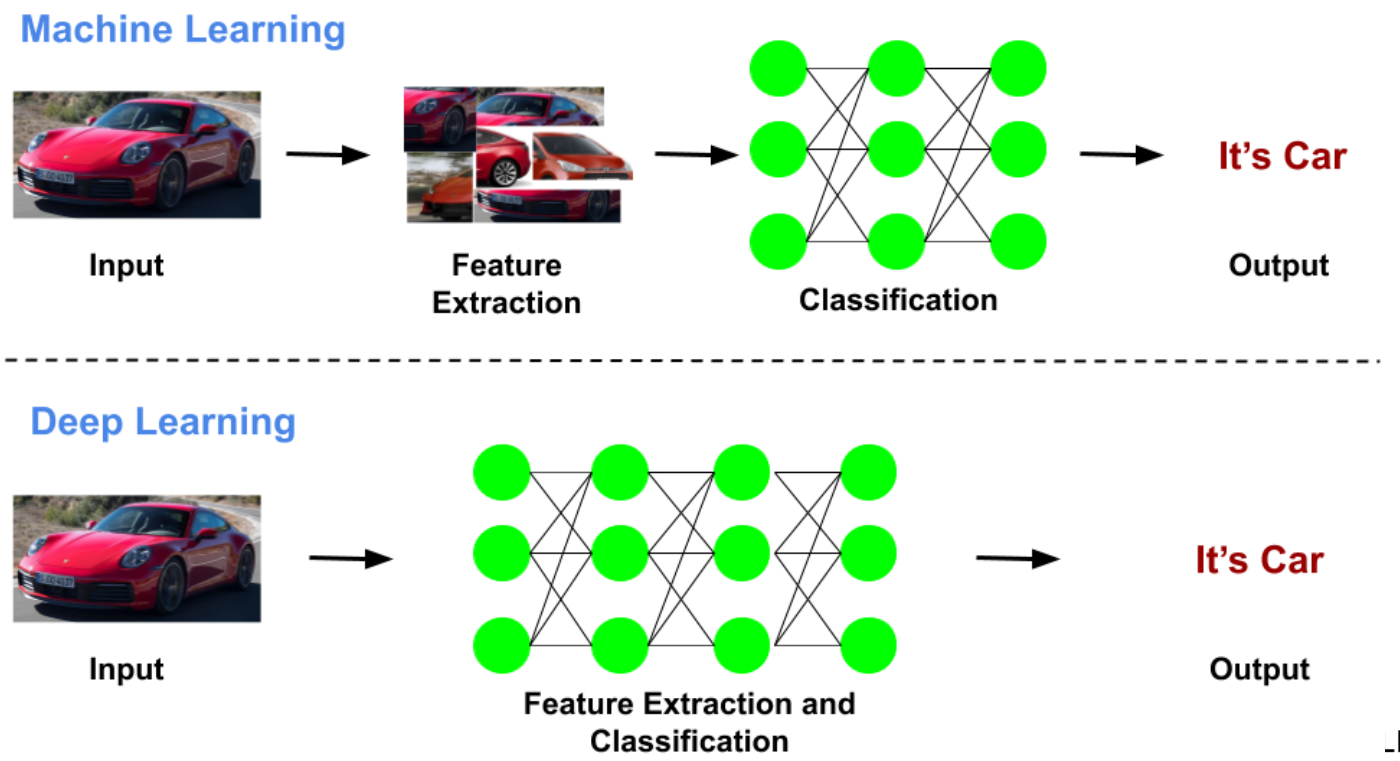


Introduction

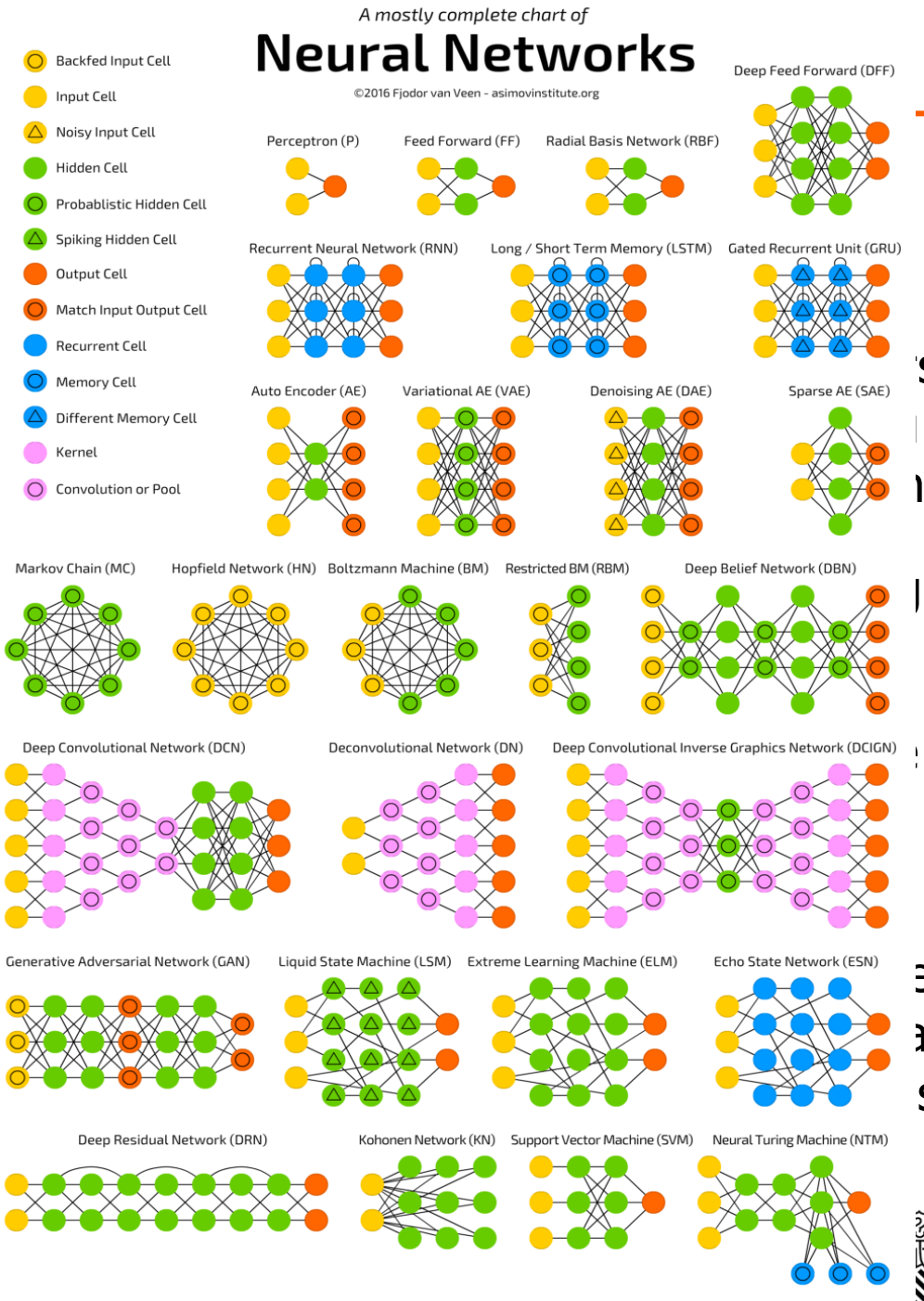
- Deep Learning follows the layer structure and the training mode (backpropagation) of ANN
- The layers extract features from the data. These models automatically perform Feature Engineering



Introduction

Some types of ANN

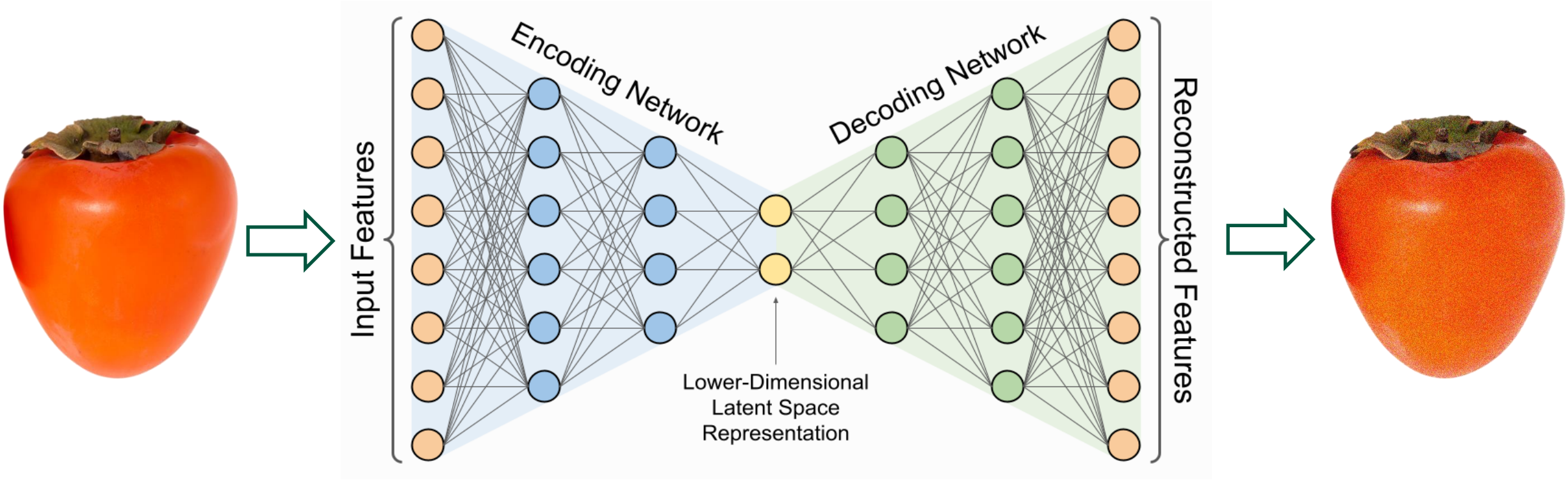
- **Multilayer Percep** ...s of neurons, where each neuron is conr ...bsequent layers. Especially useful for ...ns.
- **Convolutional Ne** ...e processing to recognize patterns ...
- **Generative Adver** ...nerate new synthetic samples from real c ... and another evaluate its quality by compa ...
- **Autoencoders**: Us ...ed data (unsupervised learning). An encod ...a, and a decoding function recreates t ...sentation.

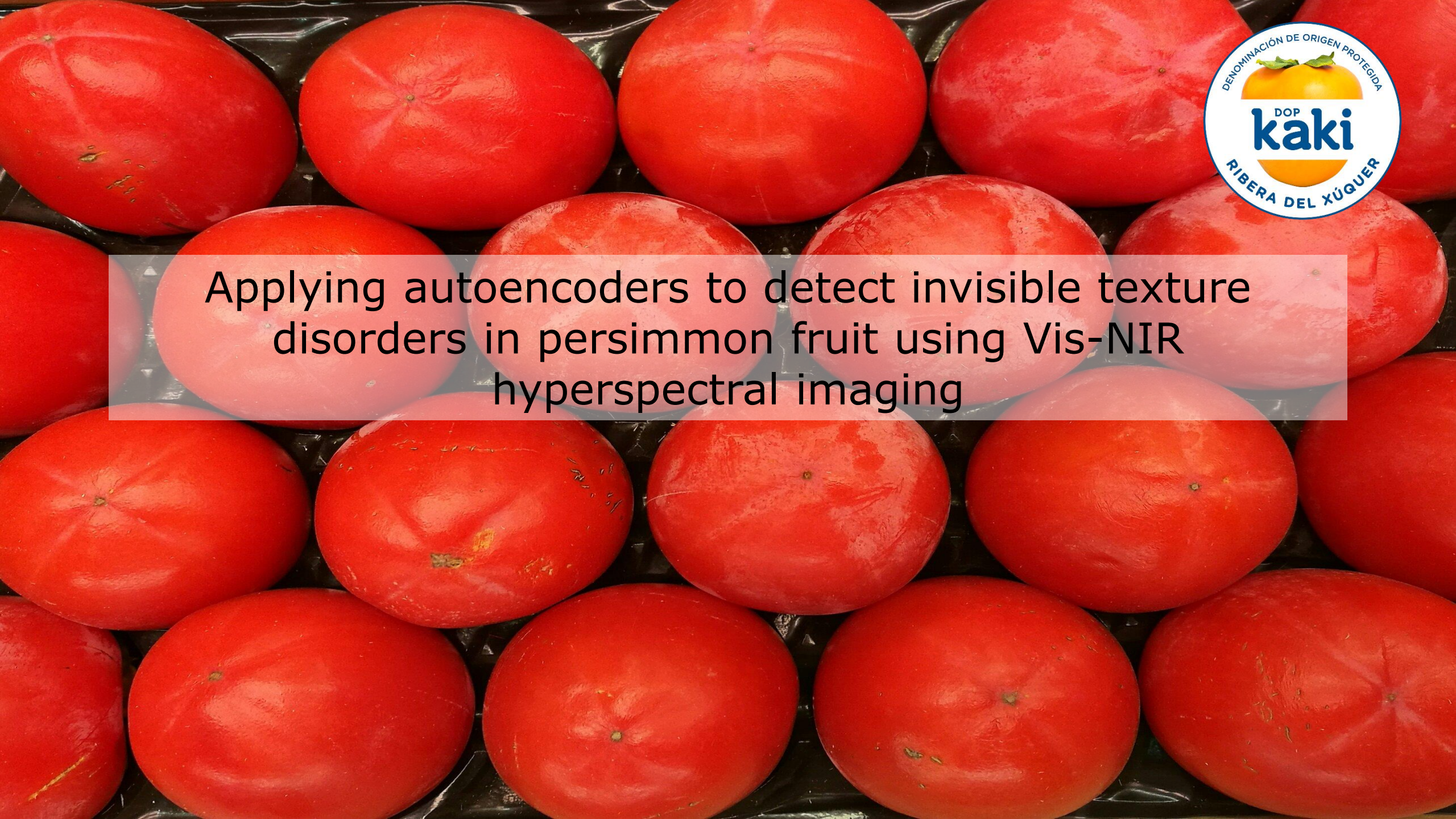


Problems using DL on sorting lines

- Models must be trained on many **(thousands) of samples**
- Such a large number of samples are **not always available**
- Sometimes it needs expensive and **complex destructive analyses** to get the ground truth from the samples
- Some quality classes are more difficult to obtain than others, resulting in a **class imbalance**
- Training is slow and requires **huge computational capacity**
- Solution: generate **artificial samples**

Autoencoders





Applying autoencoders to detect invisible texture disorders in persimmon fruit using Vis-NIR hyperspectral imaging

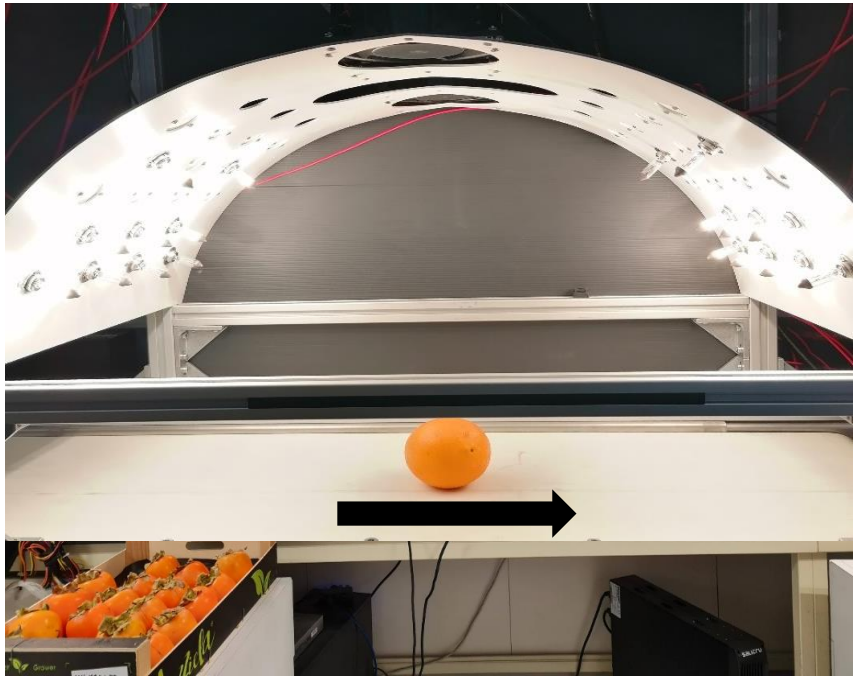
Detection of texture changes in persimmon

- Problem:
  - Persimmon: fruit that suffers undesired changes due to unappropriated storage conditions
  - Problems: fruit handling, consumer acceptance
  - Producers search for excellence
  - **Not detectable in sorting lines**

- Objectives:
  - Prove the use of autoencoders for effective feature extraction of texture parameters
  - Use hyperspectral imaging to predict texture classes

Detection of texture changes in persimmon

Image acquisition

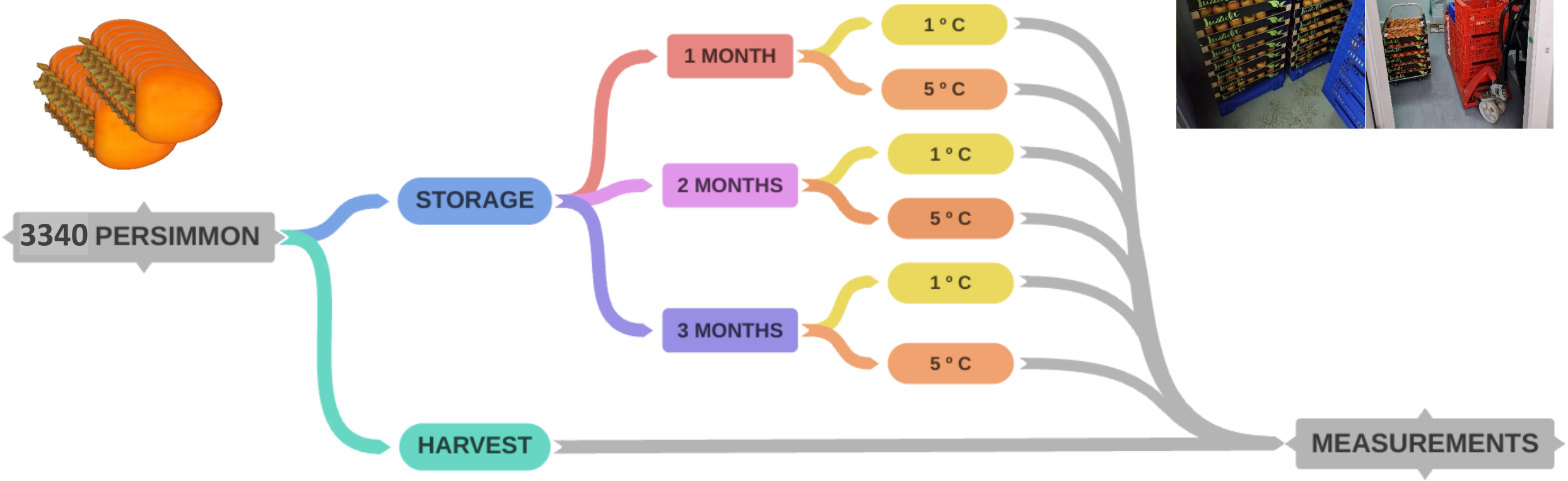


- Vis-NIR system: 420-1010 nm
- 3340 images

Texture measurement

- Instron (curves up to break down)

Materials and methods



(70% for training, 30% for validation)

Detection of texture changes in persimmon

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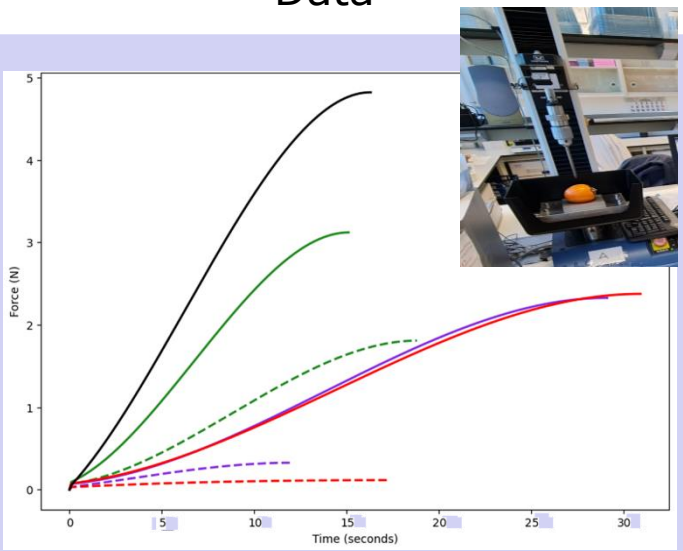
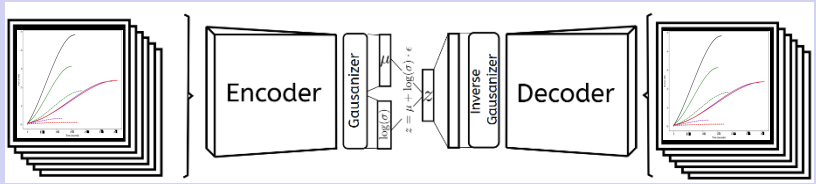
Data

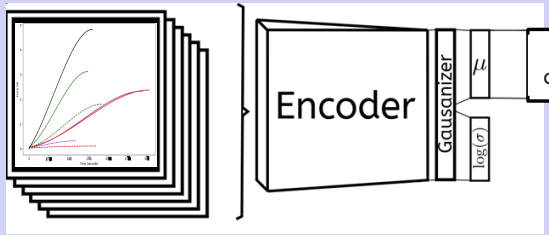


Texture curves

Autoencoder training

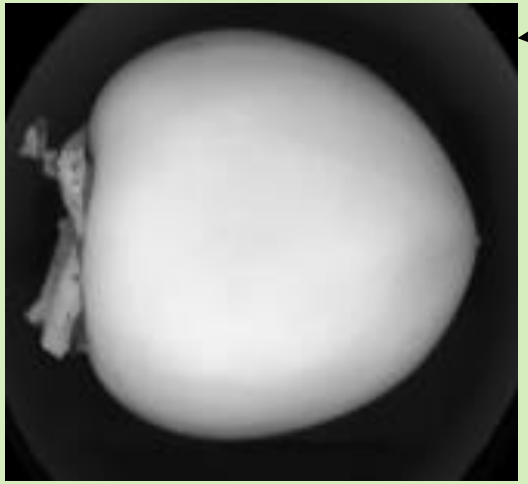


Feature extraction and classification

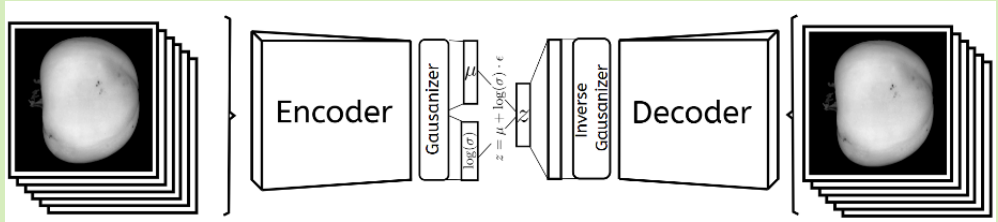


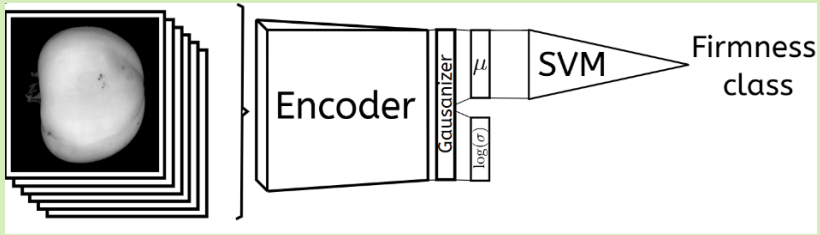
K-Means clustering → Firmness class

Sample clustering and labelling
(Result: 3 clusters)



Images and labels





Label prediction

Detection of texture changes in persimmon

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Results

- Classification in three groups of totally invisible, but extremely important, internal texture
- 99 % Data reduction using autoencoders
- 82 % Overall success

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Summary

- This is an example of the use of autoencoders to get invisible quality traits of a fruit
- Other DL techniques are also promising:
  - **DCNN** for processing colour (or multispectral) images
  - **GAN** to generate synthetic samples when they are challenging to obtain
  - **YOLO** to detect objects (i.e. yield prediction) or fruit defects

Yes: Phenotyping Sensors Provide a Potential for Sorting Lines!

Thank you for attending

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